

Unemployment Insurance

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Unemployment insurance (UI)

UI in the US:

- Eligibility rules
- Replacement rate $\approx 50\%$

→ benefits = 50% of past wage
(+ cap on benefits)

- Benefits have finite duration

→ usual duration = 26 weeks

- Duration of benefits is countercyclical

→ duration of benefits is extended when

unemployment $\uparrow$

- state $u > 6.5\%$ duration of UI benefits $\uparrow$ to 39 weeks
- state $u > 8\%$ duration $\uparrow$ to 46 weeks
- additional federal extensions

Introducing UI into matching model

- One-period model

- All workers are initially unemployed
- Size of labor force $H = 1$
- Unemployed workers search with effort $E > 0$
- Aggregate search effort = # unemployed workers $\times$ (effort / worker)
$$= E \times 1$$
$$= E$$
- Firms post V vacancies to recruit workers
- Matching function gives # of worker-firm matches $m(E, V)$

- Labor market tightness is $\theta = V/E$.
- Probability to fill a vacancy: $q(\theta)$
- Probability to find a job / unit of effort $f(\theta)$
 $\hookrightarrow$ probability to find a job $E \times f(\theta)$

Labor demand

- One representative firm.
- L workers
 - N producers
 - R recruiters
- production function: $Y = a N^\alpha$
 - wage function: $W = W(a, UI)$
 - recruiter - producer ratio: $\tau(\theta) = R/N$

$$L \text{ hires} \rightarrow V = L / q(\theta)$$

$$\rightarrow R = r \times V = r \times \frac{L}{q(\theta)}$$

$$\text{So } \frac{R}{N} = r \times \frac{L}{q(\theta)} = \frac{r}{q(\theta)} \times \left(\frac{R}{N} + 1 \right)$$

$$\tau = \frac{r}{q(\theta)} (1 + \tau)$$

$$\tau = \frac{r/q(\theta)}{1 - r/q(\theta)}$$

$$\tau(\theta) = \frac{r}{q(\theta) - r}$$

Prof. 1

$$\pi = y - w \times L$$

$$\tau(\theta) \times N = R$$

$$\pi = a \cdot N^\alpha - w \times [1 + \tau(\theta)] \times N$$

$\Rightarrow$ same as in usual model

same labor demand.

$$L^d(\theta, UI) = \left[\frac{a \alpha}{w(a, UI) [1 + \tau(\theta)]^\alpha} \right]^{1/(1-\alpha)}$$

- downward-sloping labor demand if $\alpha < 1$
- but horizontal labor demand if $\alpha = 1$
- L^d responds to UI if w does

Representative worker

- employed worker: consume C^e
- unemployed worker: consume $0 < C^u < C^e$
- $\rightarrow$ gap between C^e & C^u is determined by UI
- generous UI system: C^u close to C^e

- nongenerous UI system: C^u much lower than C^e

utility function of workers

- consumption utility $U(C)$: increasing, concave (risk-averse workers; value insurance)
- disutility from job search $\psi(E)$ increasing, convex - Quadratic disutility $\psi(E) = E^2/2$.
- generosity of UI is well-measured by the utility gain from work: $\Delta U = U(C^e) - U(C^u)$
- $\Delta U > 0$
- UI generous $\Rightarrow \Delta U$ is low
- UI nongenerous $\Rightarrow \Delta U$ is high
- increase generosity of UI reduce ΔU .

Worker's problem maximize expected utility by choosing search effort E

$$\max_E U(C^u) + E f(\theta) \Delta U - E^2/2$$

Concave function $\rightarrow$ first-order condition gives

global maximum -

take derivative of objective function

$$f(\theta) \Delta U - E = 0$$

effort chosen by workers

$$E^s(\theta, UI) = f(\theta) \Delta U$$

- $UI \downarrow \Rightarrow$ gain from working $\uparrow \Rightarrow$ incentive to search $\uparrow \Rightarrow E \uparrow$

$$\partial E^s / \partial UI < 0$$

- $\theta \uparrow \Rightarrow$ return on effort $\uparrow \Rightarrow$ incentive to search $\uparrow \Rightarrow E \uparrow$

$$\partial E^s / \partial \theta > 0$$

Labour supply

$$L^s(\theta, UI) = E^s(\theta, UI) \times f(\theta)$$

$$\theta = 0 \Rightarrow f(\theta) = 0 \Rightarrow L^s = 0$$

$$\partial L^s / \partial UI < 0 \quad \cdot \quad UI \text{ decreases, labour supply}$$

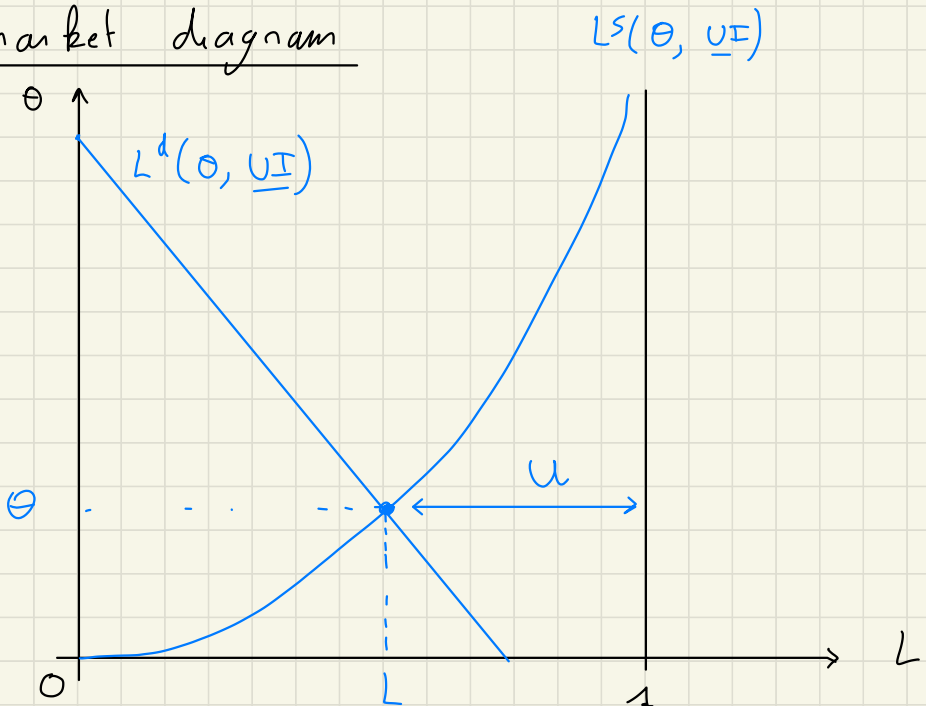
$$- \partial L^S / \partial \theta > 0$$

Labour market equilibrium with UI

$$L^S(\theta, UI) = L^d(\theta, UI)$$

Implicitly, θ is function of UI $\theta(UI)$

Labour market diagram

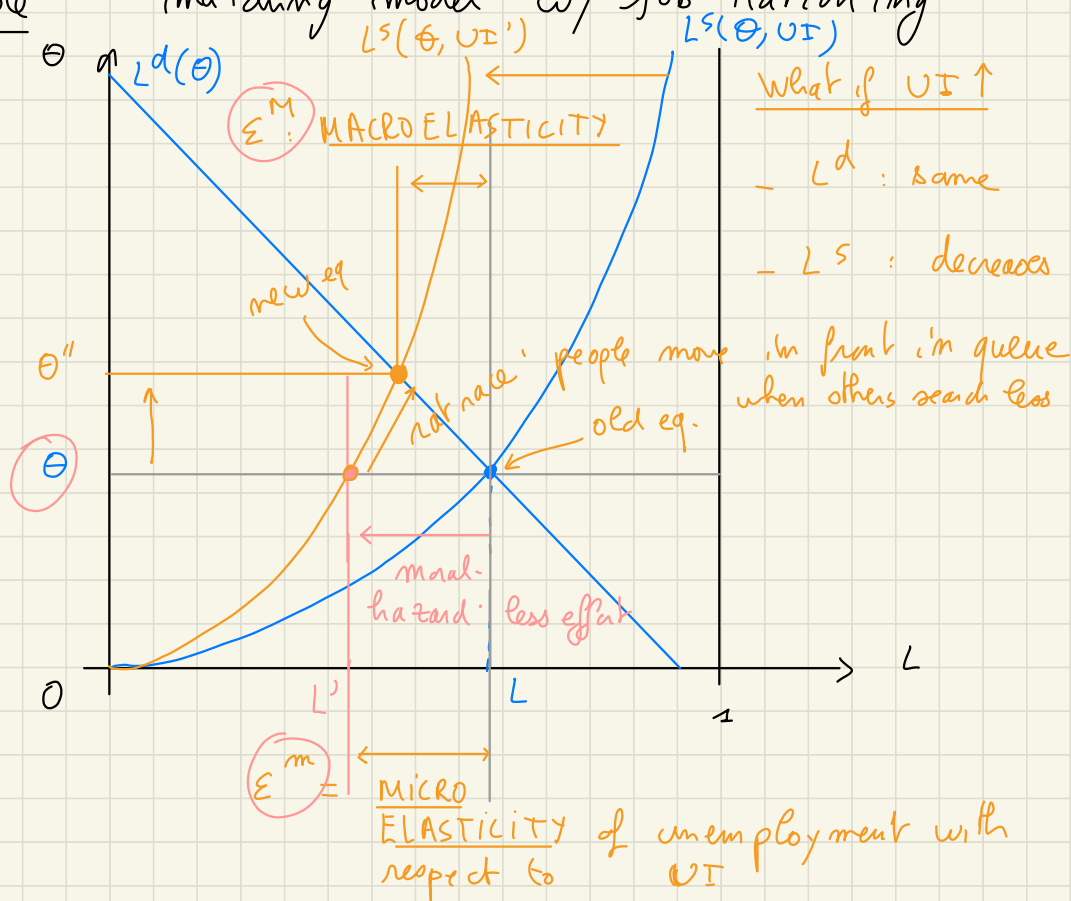


Effects of UI on labor market

1) Wages do not respond to UI (most realistic case)

+ concave production function (ie a downward sloping labor demand)

Example matching model w/ job rationing

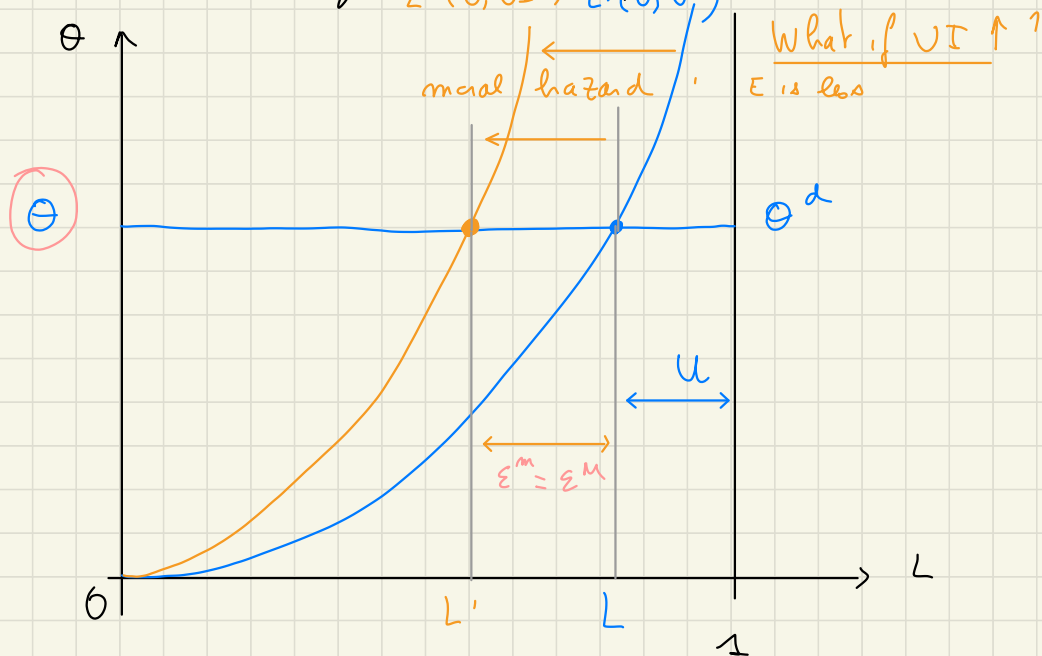


Effects of an increase in UI .

- $L \downarrow$, $u \uparrow$
- $\theta \uparrow$
- $E \downarrow$ MACRO MICRO
- $0 < \epsilon^m < \epsilon^M$

2) Wages do not respond to VI + production function is linear (i.e. the labor demand is horizontal)

Example matching model w/ rigid wages



Effects of an increase in VI.

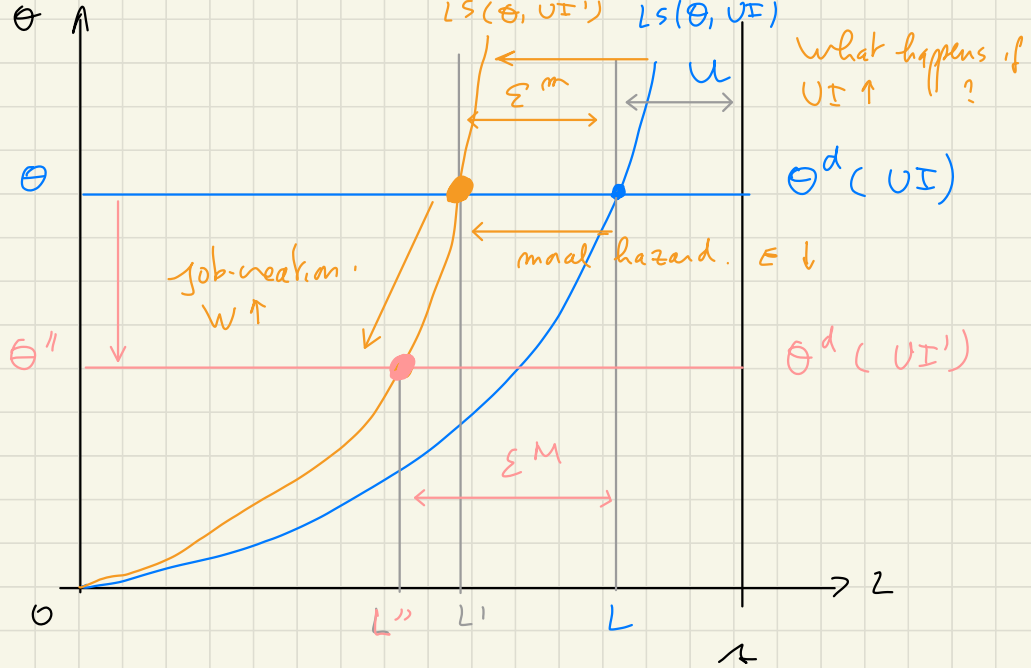
- $L \downarrow$ & $u \uparrow$

- $\theta \rightarrow$

- $E \downarrow$

- $0 < \varepsilon^u = \varepsilon^m$

3) Wages increase with VI (bargaining) & linear production function (i.e. horizontal L^d)



Effects of higher UI.

- $L \downarrow$ $u \uparrow$
- $\theta \downarrow$
- $\epsilon \downarrow$
- $W \uparrow$
- $0 < \epsilon^m < \epsilon^M$

Optimal UI.

Social welfare is .

$$SW = L \cdot U(c^e) + (1-L) \cdot U(c^u) - \frac{E^2}{2}$$

$\uparrow$
 $\psi(E)$

Social planner chooses ψ to maximize

SW subjects to the following constraints:

- budget constraint for government ($\Rightarrow$) resource constraint in economy.

$$L c^e + (1-L) c^u = Y = a \cdot N^2$$

$\nearrow$
total consumption

- workers response :
 - $E = E^s(\theta, \underline{UI})$
 - $L = L^s(\theta, \underline{UI})$
- equilibrium response :
 - $\theta = \theta(\underline{UI})$
 - given by $L^d(\underline{\theta}, \underline{UI}) \nearrow L^s(\underline{\theta}, \underline{UI})$
- Solving social planner's problem
 - * All variables in social planner's problem can be expressed as function of $(\theta, \underline{UI})$

* Social welfare can be expressed as function of (θ, UI)

* Social planner's problem becomes

$$\max_{UI} SW(\theta(UI), UI)$$

Optimal UI is given by first-order condition

$$\frac{dSW}{dUI} = 0 \Rightarrow 0 = \underbrace{\frac{\partial SW}{\partial UI} \Big|_{\theta}}_{\text{DAILY-CHETTY FORMULA}} + \underbrace{\frac{\partial SW}{\partial \theta} \Big|_{UI} \cdot \frac{d\theta}{dUI}}_{\text{CORRECTION TERM}}$$

$$\textcircled{1} \frac{\partial SW}{\partial UI} \Big|_{\theta} = 0$$

UI that maximizes welfare, keeping θ constant

→ optimal UI in a "partial equilibrium" setup or "micro" setup

→ UI solving optimally tradeoff b/w incentives & insurance

→ UI given by a

public-finance formula called "Barry-chetty formula".

Formula gives optimal UI as a function of 2 statistics.

- ϵ^m microelasticity of unemployment

incentive cost of UI $\uparrow$

$w \rightarrow t \rightarrow UI$
 $\epsilon^m \uparrow$

$\Rightarrow$ optimal UI $\downarrow$

- $U'(c^e) / U'(c^u)$ ratio of marginal utilities, measuring need for insurance

$E[0,1] \sim U'(c^e) / U'(c^u) \uparrow \Rightarrow$ optimal UI $\downarrow$

insurance value of UI $\downarrow$

$$\textcircled{2} \frac{\partial SW}{\partial \theta} \Big|_{UI}$$

efficiency term captures whether the labor market operates efficiently or not -

Three possible cases

a) $\frac{\partial SW}{\partial \theta} = 0$: labor market tightness is efficient

→ Barly-chetty remains valid

b) $\frac{\partial SW}{\partial \theta} > 0$: labor market tightness is inefficiently low → labor market is inefficiently slack

→ Barly-chetty formula not valid anymore

c) $\frac{\partial SW}{\partial \theta} < 0$: tightness is inefficiently high → labor market is inefficiently tight.

→ Barly-chetty formula is not valid anymore

③ $d\theta/dUI$

Effect of UI on equilibrium tightness

a) $d\theta/dUI = 0$: UI has no effect on tightness

- happens in matching model
w/ rigid wage
- $\varepsilon^m = \varepsilon^m$

→ Barly-Chetty formula remains valid

b) $d\theta/dv\pi > 0$. $\theta \uparrow$ when $v\pi \uparrow$

- happens in matching model w/
job rationing

- $0 < \varepsilon^m < \varepsilon^m$

→ Barly-Chetty formula has to be corrected

(A) if labor market is inefficiently tight

(boom): correction term < 0 so

optimal $v\pi$ is less than in Barly-Chetty formula.

(B) if labor market is inefficiently slack

(plump): correction term > 0 so

optimal $v\pi$ is more than in Barly-Chetty formula.

⇒ optimal $v\pi$ is countercyclical

$\Rightarrow$ optimal UI is more generous in slumps than in booms (as in US)

c) $d\theta/dUI < 0$ $\theta \downarrow$ when $UI \uparrow$

- happens in standard matching model (bargaining + linear production function)

- $0 < \varepsilon^m < \varepsilon^n$

$\rightarrow$ Barro - chetty formula has to be corrected

$\Rightarrow$ Optimal UI is procyclical

$\Rightarrow$ Optimal UI is more generous in booms than in slumps (opposite of US policy)